

Technology Stack

Date	7 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask is lightweight and efficiently handles user requests, processes input data, and communicates with the machine learning model.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MySQL	Stores applicant information, prediction history, and user records securely with efficient data management.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render (or Railway)	Provides simple, reliable, and cost-effective deployment for the web application.

5	Version Control & CI/CD	GitHub	Enables version control, collaboration, code backup, and project management.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib	Scikit-learn trains the prediction model, Pandas and NumPy process the dataset, and Joblib saves and loads the trained model efficiently.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		